WORKED SOLUTION

STEP III 2010, Question 7

Pure • Maclaurin Series

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s3-q7

The key idea

Squaring away the awkward square root is what makes everything work: once we write $\sqrt{1-x^2}\,y'$ as a single tidy expression and differentiate again, the second-order equation falls out. After that, repeated differentiation produces a pattern in the coefficients, and induction proves it. Putting $x = 0$ into the general relation turns it into a recurrence for the Maclaurin coefficients, and for even m that recurrence hits a factor of zero, so the series stops.

Solution

The differential equation

Write $\theta = \arcsin x$, so $\sin\theta = x$ with $|\theta| < \frac{1}{2}\pi$, and $\cos\theta = \sqrt{1-x^2} > 0$ for $|x| < 1$. Then $\dfrac{\mathrm{d}\theta}{\mathrm{d}x} = \dfrac{1}{\sqrt{1-x^2}}$ and $y = \cos m\theta$, so by the chain rule

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{m\sin m\theta}{\sqrt{1-x^2}}, \qquad \text{that is,} \qquad \sqrt{1-x^2}\,\frac{\mathrm{d}y}{\mathrm{d}x} = -m\sin m\theta.$$

Differentiate this with respect to x, using the product rule on the left:

$$\sqrt{1-x^2}\,\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{x}{\sqrt{1-x^2}}\,\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{m^2\cos m\theta}{\sqrt{1-x^2}} = -\frac{m^2y}{\sqrt{1-x^2}}.$$

Multiplying through by $\sqrt{1-x^2}$ (which is non-zero) gives

$$(1-x^2)\,y'' - x\,y' + m^2y = 0,$$

as required. (Here and below $y^{(n)}$ means $\mathrm{d}^ny/\mathrm{d}x^n$.)

The third- and fourth-order equations

Differentiate the equation just proved. The product $(1-x^2)y''$ gives $(1-x^2)y''' - 2xy''$ and xy' gives $xy'' + y'$, so

$$(1-x^2)\,y''' - 3x\,y'' + (m^2-1)\,y' = 0.$$

Differentiating once more in the same way,

$$(1-x^2)\,y^{(4)} - 2x\,y''' - 3x\,y''' - 3y'' + (m^2-1)\,y'' = 0,$$

so

$$(1-x^2)\,y^{(4)} - 5x\,y''' + (m^2-4)\,y'' = 0.$$

The general relation

The coefficients of x run $1, 3, 5$ and the constants run $m^2-0^2,\ m^2-1^2,\ m^2-2^2$, so we conjecture that for every integer $n \geq 0$

$$(1-x^2)\,y^{(n+2)} - (2n+1)\,x\,y^{(n+1)} + (m^2-n^2)\,y^{(n)} = 0. \qquad (*)$$

We prove this by induction on n. The case $n=0$ is the equation proved first. Suppose $(*)$ holds for some n. Differentiating it,

$$\begin{aligned}(1-x^2)\,y^{(n+3)} - 2x\,y^{(n+2)} - (2n+1)\,x\,y^{(n+2)} \\ - (2n+1)\,y^{(n+1)} + (m^2-n^2)\,y^{(n+1)} = 0.\end{aligned}$$

Collecting terms, the coefficient of $x\,y^{(n+2)}$ is $-(2n+3) = -(2(n+1)+1)$ and the coefficient of $y^{(n+1)}$ is $m^2-(n^2+2n+1) = m^2-(n+1)^2$. So

$$(1-x^2)\,y^{(n+3)} - (2(n+1)+1)x\,y^{(n+2)} + (m^2-(n+1)^2)y^{(n+1)} = 0,$$

which is $(*)$ with $n+1$ in place of n. By induction, $(*)$ holds for all $n \geq 0$.

The Maclaurin series

At $x=0$ we have $\theta = 0$, so $y(0) = \cos 0 = 1$ and $y'(0) = -m\sin 0 = 0$. Putting $x=0$ in $(*)$ kills the middle term:

$$y^{(n+2)}(0) = (n^2-m^2)\,y^{(n)}(0).$$

Since $y'(0)=0$, every odd derivative vanishes at 0. For the even ones,

$$\begin{aligned}y''(0) &= (0-m^2)\cdot 1 = -m^2,\\ y^{(4)}(0) &= (2^2-m^2)(-m^2) = m^2(m^2-2^2),\\ y^{(6)}(0) &= (4^2-m^2)\,m^2(m^2-2^2) = -m^2(m^2-2^2)(m^2-4^2),\end{aligned}$$

and in general $y^{(2k)}(0) = (-1)^k m^2(m^2-2^2)(m^2-4^2)\cdots(m^2-(2k-2)^2)$. Hence the Maclaurin series is

$$y = 1 - \frac{m^2x^2}{2!} + \frac{m^2(m^2-2^2)\,x^4}{4!} - \frac{m^2(m^2-2^2)(m^2-4^2)\,x^6}{6!} + \cdots,$$

and the first three non-zero terms are $1 - \dfrac{m^2x^2}{2!} + \dfrac{m^2(m^2-2^2)x^4}{4!}$. (These are all non-zero provided $m \neq 0, \pm 2$; for those values the series has fewer terms, as the next part explains.)

Even m

Let m be an even integer, and write $|m| = 2p$. The factor $m^2 - (2p)^2$ is zero, and it appears in $y^{(2k)}(0)$ as soon as $2k - 2 \geq 2p$. So every derivative of order greater than $|m|$ is zero at $x = 0$, and the Maclaurin series stops at the $x^{|m|}$ term.

Ask yourself A series that stops at $x = 0$ only tells us about derivatives at one point. How do we know y really *equals* this polynomial for every $|x| < 1$?

We use $(*)$ again, now on the whole interval. Take $n = |m|$, so that $m^2 - n^2 = 0$, and put $v = y^{(n+1)}$. Then $(*)$ reads

$$(1 - x^2)\, v' - (2n+1)\, x\, v = 0 \qquad (|x| < 1).$$

This first-order equation has integrating factor $(1-x^2)^{(2n-1)/2}$; indeed

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big[(1-x^2)^{(2n+1)/2}\, v\Big] = (1-x^2)^{(2n-1)/2}\Big[(1-x^2)\, v' - (2n+1)\, x\, v\Big] = 0,$$

so $(1-x^2)^{(2n+1)/2}\, v$ is constant on $(-1, 1)$. Its value at $x = 0$ is $v(0) = y^{(n+1)}(0) = 0$, because $n+1$ is odd. Hence $v \equiv 0$, that is, $y^{(|m|+1)}(x) = 0$ for all $|x| < 1$, and so y is a polynomial of degree at most $|m|$. A polynomial equals its own Maclaurin series, so for $|x| < 1$

$$\cos(m \arcsin x) = 1 - \frac{m^2x^2}{2!} + \frac{m^2(m^2-2^2)\, x^4}{4!} - \cdots,$$

the right-hand side being the terminating series above.

Now let $|\theta| < \frac{1}{2}\pi$ and put $x = \sin\theta$. Then $|x| < 1$ and $\arcsin(\sin\theta) = \theta$ (this is exactly where the restriction on θ is used), so

$$\cos m\theta = 1 - \frac{m^2 \sin^2\theta}{2!} + \frac{m^2(m^2-2^2)\sin^4\theta}{4!} - \cdots,$$

a polynomial in $\sin\theta$ beginning as required.

Finally, the coefficient of $\sin^{|m|}\theta$ is $y^{(|m|)}(0)/|m|!$, which is $\pm\, m^2(m^2-2^2)\cdots(m^2-(|m|-2)^2)/|m|!$. None of these factors is zero, since $(2j)^2 < m^2$ for $2j \leq |m| - 2$. So for $m \neq 0$ the polynomial has degree exactly $|m|$ (that is, degree m when m is a positive even integer); for $m = 0$ it is the constant 1.

Checks

- $m = 2$: the series gives $1 - 2\sin^2\theta$, which is the familiar $\cos 2\theta$.

- $m = 4$: the series gives $1 - 8\sin^2\theta + \frac{16\cdot 12}{24}\sin^4\theta = 1 - 8\sin^2\theta + 8\sin^4\theta$. Directly, $\cos 4\theta = 1 - 2\sin^2 2\theta = 1 - 8\sin^2\theta\cos^2\theta = 1 - 8\sin^2\theta + 8\sin^4\theta$. They agree.
- We checked $(*)$ numerically for $n = 0, \ldots, 8$ at 200 random pairs (m, x), using Taylor expansions of $\cos(m \arcsin x)$ about each point (largest relative residual about 10^{-13}), and checked $\cos m\theta$ against the polynomial for every even m up to 12. For example, at $m = 6$, $\theta = 1$ both sides equal 0.96017.

Common slips

- Differentiating $(1 - x^2)y^{(n+2)}$ and forgetting the $-2x\,y^{(n+2)}$ term from the product rule: this is what turns $2n + 1$ into $2n + 3$.
- Assuming that a Maclaurin series which stops must equal the function. Here it does, but it needs the argument with $v = y^{(|m|+1)}$ (or another justification).
- Giving the degree as $m/2$: the polynomial contains only even powers of $\sin\theta$, but the highest power is $\sin^m\theta$.

Beyond STEP

Substituting $x = \cos\phi$ instead turns the same kind of equation into Chebyshev's equation $(1 - x^2)y'' - xy' + n^2y = 0$, whose polynomial solutions $T_n(\cos\phi) = \cos n\phi$ are the Chebyshev polynomials, central in numerical approximation. Our trick of putting $x = 0$ into the differentiated equation to get a recurrence for coefficients is the start of the *power series method* for linear ODEs, where a series terminating for special parameter values is how Legendre, Hermite and Chebyshev polynomials all arise. Boyce and DiPrima, *Elementary Differential Equations*, has a clear chapter on series solutions.

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